Biospectra : Vol.9(1), March, 2014. pp 225-230
An International Biannual Refereed Journal of Life Sciences

Plant Sciences

ISSN : 0973-7057

Ethnobotanical investigations on anti-diabetic plants used by tribal inhabitants of Hazaribagh of Jharkhand, India

Rajesh Sinha[a]* & Radha Krishan Jha[a]
*[a]Department of Botany, Ranchi University, Ranchi, Jharkhand

Received , 18th January, 2014 ;Revised: 20th February, 2014

Abstract : An ethnobotanical studies was conducted to collect information about medicinal plants used for the treatment of diabetics and associated complications by tribals in Bishnugarh, Hazaribagh District, Jharkhand, India. The indigenous knowledge of local traditional healers and tribal healers of Bishnugarh for native plants used for the treatment of diabetics related health disorders were collected through questionnaire and personal interviews. The present study describes the identification of plants and documentation of plants well known for medicinal value by the virtue of tribal traditional practices. A total of 19 informants within the age group of 42 to 70 were interviewed, among them four were herbal practitioners. Our investigation revealed that, the local medico-botanical knowledge remains pretty specific in the area and the traditional healers, tribal inhabitants use 53 species of plants belonging to 38 families to treat diabetics and related complications. Anti-diabetic medicinal plants used by different tribes have been listed along with preparation of their medicine. This paper showed many medicinal plants from different families of which are purely anti-diabetic potential.

Key Words: Ethnomedicine, Anti-diabetic Plants, Ethnobotanical, Traditional Knowledge, Bishnugarh.

INTRODUCTION

Ethnobotanical studies of traditional herbal remedies used for diabetes around the world have identified more than 1,200 species of plants with hypoglycemic activity.[1] In India, indigenous remedies have been used in the treatment of DM since the time of Charaka and Sushruta (6th century BC).[2] Ayurveda and other Indian traditional medicine use plants in treatment of diabetics.[3] However, traditional knowledge, derived empirically, has to be supported by scientific testing.

Several herbs have shown anti-diabetic activity when assessed with the available experimental techniques.[4] Wide array of plant derived compounds with consistent antidiabetic activity have proven their possible use in the treatment of DM.[5] Plants have shown varying degree of hypoglycemic and anti-hyperglycemic activity.[2]

*Corresponding author :
Phone : 09431382649
E-mail : rajeshsinha912@gmail.com

Traditional plant therapies and attention to dietary measures increasing in recent decades has ignited a new wave of research interest in plant kingdom. India holds a credibility of diverse social, cultural and medical heritage with an unbroken tradition coming down across millennia. Though, medical heritage is centuries old, millions people in rural area still depend on traditional medicine to congregate their healthcare needs.[6] A mounting body of research is hoisting the credibility of traditional knowledge in meeting the challenges of primary healthcare and management of natural resource globally.[7] Diabetes mellitus (DM) is a metabolic disorder caused by insufficient or inefficient insulin secretary response and it is characterized by increased blood glucose levels (hyperglycemia).[8] The frequency of this disorder is on the rise globally, is likely to hit 300 million by 2025 with India projected to have the largest number of diabetic cases.[9] The Indian traditional system of medicine prescribed plant therapies for diseases including diabetes mellitus called madhumeh in Sanskrit.

Considerable interest for ethnobotanical community medications as they are recognized to contain valuable medicinal properties in different parts of the plant and a number of plants have shown varying degree of hypoglycemic and anti-hyperglycemic activity.[10] Experimentally, Shanmugasundaram.[11] confirmed the efficacy of traditional preparations claimed to be effective in the treatment of diabetics. Wide array of plant derived compounds with consistent antidiabetic activity have proven their possible use in the treatment of DM.[5] The tribals of Bishnugarh block of Hazaribagh District, Jharkhand are Shanthal, Ho, Oraon and Bhuian and they have good knowledge of medicinal plants, diseases and treatment modalities.

Decoction, powder, paste, infusion and in crude form, with other additives like honey, cow milk, cattle milk, to relieve from different diabetic ailments. Documenting indigenous knowledge is important from the view point of conservation of biological resources and their sustainable utilization in the management of diabetics and its related complications. Therefore, present investigation on the use of ethnomedicinal plants for the treatment of diabetics has been taken up in Bishnugarh block of Hazaribagh district, Jharkhand, India.

MATERIALS AND METHODS

Ethnobotanical survey

Field investigations were conducted in Bishnugarh block of (23.98°N 85.35° E), Hazaribagh District,Jharkhand. In the present study, the methods suggested by Jain and Goel (1995) were adopted to survey the ethno botanical uses of medicinal plants,[4] specimen collection, field Notes, processing of the material was followed according to the Standard protocol suggested by BSI. There were 19 informants with in the age 42 – 70 of different tribal groups. Among them four were tribal practitioner. The ethnobotanical data were collected using questionnaire, interviews and discussions in their local dialect.

RESULTS AND DISCUSSION

In this study, we focused mainly on plant species reported by the local people in and around the study area for their medicinal uses. Present data are the general results of the ethnobotanical survey conducted from February to July (2013). In the present investigation 53 medicinal plants are used for the treatment of diabetes (Table 1). Present investigation indicates that is blessed with splendid diversity of ethnomedicinal plants. Results depict that fresh plant materials were invariably preferred for the treatment of long term complications associated with diabetics. Field visits in at different areas of Bishnugarh, Hazaribagh District, Jharkhand has been done. At the same time, we explored the forest around each village, and local experts showed us the plants known to them and described preparations of diabetic antidotes. Plant names were transcribed phonetically, mode of preparation of their medications with dosage,usage and prescribed medicinal plants by each specific ethnomedicinal practioners of different tribes such as Shanthal, Ho, Oraon and Bhuian and common herbal practioners in Bishnugarh block of Hazaribagh, Jharkhand have been documented in Table 1. During this prescribed medications and administration, eating sugar containing food stuffs are totally discouraged.. By using an empirical approach where the same specimens were shown in Bishnugarh block area, we obtained replicate data sets. Ethnic people identified diabetics with some ailments like polyurea, smell in urine, and gathering of ants where urine is passed. Our investigations claimed comparative data and also potential source for raw material for pharmaceutical products.

The tribal inhabitants revealed medicinal importance of 53 species belonging to 38 families have been studied in relation to diabetes and their complications should be studied further to find the efficacy and to prove scientifically in international standards for the benefit of the common man and to show the role of ethnomedicine as new vista for the treatment of diabetes is a practical, cost-effective and biologically safe. Furthermore experiments are carrying out with medicinal plants belonging to different families as mentioned above. More investigations must be carried out to evaluate the mechanism of action of medicinal plants with anti- diabetic effect. The toxic effect of these plants should also be elucidated.

Therefore active research on identification and isolation of anti-diabetic compounds from plants, their pyhtochemical studies as well as developing new areas where the likelihoods may be increased are in great need.

Further, The present investigation deep-rooted the benefits and traditional uses of medicinal plants with hypoglycemic effects and established some pharmacological evidence to support the folklore claim that it is used in the management and remedial of diabetes.

Table 1:Tabulation showing Anti diabetic medicinal plant species used by the inhabitants of Bishnugarh block area of Hazaribagh, Jharkhand

Sl. No	Botanical Name	Family	Vernacular	Ethnobotanical Usage
1.	*Annona reticulata* (L).	Annonaceae	Ramphal	Leaf juice mixed with 150 ml boiled cow milk and given two times daily upto five months
2	Asparagus racemosus Willd.	Asparagaceae	Shatavari	The filtered extract is given along with 5g of jaggery daily for two months.
3.	*Acacia arabica* (Lam.) Willd	Mimosaceae	Babul	Seeds hypoglycemic
4.	Achyranthes aspera L.	Amaranthaceae	Pratyakpushpi	Dried leaf powder (2-3g) is given with water. Five fresh leaves taken daily in the morning for four weeks.
5.	*Aegle marmelos* (L.) Corr	Rutaceae	Bael	Leaf powder with cow's milk is taken orally.
6.	Aloe vera (L.) Burm.f.	Asparagaceae	Musaber	Succulent stems (gel) are processed and taken with floral nectar and bee honey early hours.
7.	*Allium sativum* Linn.	Alliaceae	Lehsun	Leaf, bulb antidiabetic taken orally.
8.	*Allium cepa* Linn	Alliaceae	Pyaj	Leaf juice with honey or milk and taken orally.
9.	*Andrographis paniculata (Burm.f.) Nees*	Acanthaceae.	Kalmegh	Leaf decoction is given with jaggery water. Seven fresh leaves are prescribed daily in the morning for 15 days.
10.	Annona reticulata L.	Annonaceae	Sharifa	Leaf juice mixed with 150 ml boiled cow milk and given two times daily upto five months.
11.	Azadirachta indica A.Juss.	Meliaceae	Neem	Leaves consumed in morning. Few seeds of the plant made into a paste with rice juice and ghee should be given after meal in case of Chronic diabetes.
12.	*Brassica juncea* (Linn.)	Bracecaceae	Rai	Seed powder with milk is taken orally
13.	*Bombax ceiba* L	Bombacaceae	Semal	Seed powder with goat's milk is taken orally.
14.	Boerhaavia diffusa Linn.	Nyctaginaceae	Punarnava	The raw leaf juice (10 ml) is used and also advised to take the leaves and tender branch tips as vegetables.
15	*Coccinia indica* W & A	Cucurbitaceae	Kanduri	Leaf/ Fruit decoction is taken topically.
16.	*Cajanus cajan* (Linn.) Millsp	Fabaceae	Arhar	Seeds cooked and taken along with food.
17.	Cinnamomum verum	Lauraceae	Dal-chini	Leaf powder (2-3gm) with one glass of milk given daily is effective within a short period.
18.	*Cajanus cajan* (Linn.)	Fabaceae	Arhar/Tuvar	Seeds cooked and taken along with food.
19.	Citrullus colocynthis (L.) Schrad.	Cucurbitaceae.	Indrayan	Fruits with mixture of pepper, salt taken at any time in daily diet.
20.	Coccinia grandis (L.) Voigt	Cucurbitaceae	Kunduru	Fresh leaves grinded mixed with boiled cattle milk
21.	*Cuminum cyminum* Linn.	Apiaceae	Jeera	Fruit soaked in water and taken orally.
22.	*Corallocarpus epigaeus (Rottler) C.B.Clarke*	Cucurbitaceae.	Aila	The fresh root used with water and made a paste and consumed orally or the fresh leaf with fruit mix in water given orally
23.	*Cucumis sativus L.*	cucurbitaceae	Kheera	Fruits taken at any time .Seeds (2g) made into paste with liquorice (Glycyrrhiza glabra) is given daily for 15 days.
24.	*Dillenia indica L.*	Dilleniaceae	Chalta	Fruit ground to chutney(Tamarindus,Capsicum, Allium sativum mixed equally and prepared) taken with meal

25.	*Erythrina indica* Lam.	Fabaceae	Pangri	Leaves consumed to treat diabetics.
26.	*Euphorbia antiquorum* Linn.	Euphorbiaceae	Tidhara	Leaves /fruits consumed to treat diabetics.
27.	*Eucalyptus globulosus St.-Lag.*	Myrtaceae.	Eucalyptus	Fresh leaves squashed mixed with Coriandrum leaf paste and taken early hours
28.	*Ficus benghalensis* Linn.	Moraceae	Bargad	Fruits made into paste mixed with jiggery taken before going to bed.
29.	*Gymnema sylvestre*	Asclepiadaceae	Gurmar	Half a teaspoon leaf powder with pinch of jaggery powder mixture taken orally was given daily for 30 days.
30.	*Hemidesmus indicus (L.)*	Apocynaceae	Magrabu	Root tonic given early morning with 2-3 tea spoonful with honey .Root decoction given in early morning
31.	*Hiptage benghalensis*	Malpighiaceae	Madhavi lata	Decoction of the Leaf extract with Allium sativum bulbs in little proportion
32.	*Hibiscus rosa-sinesis* Linn.	Malvaceae	Jasum	Tender fresh leaves used to cure diabetes.
33.	*Lantana camara* Linn.	Verbenaceae	Raimuniya	Leaf and fruits consumed raw - treat diabetics
34.	*Lagerstroemia speciosa(L.)*	Lythraceae	Jarul	Bark dissolved in water in full night ,taken with empty stomach
35.	*Litsea glutinosa (*Lour.)	Lauraceae	Maida lakri	Fruits (unripe and riped in 1:1 gm) crushed made into paste consumed with honey or jagerry .
36.	*Momordica charantia* L .	Cucurbitaceae	Karela	Fruit juice is taken to treat diabetics
37.	. *Marsilea minuta*	Marsileaceae	Chaupatiya	Leaf juice is used for diabetes.
38.	*Mangifera indica L.*	Anacardiaceae	Aam	Leafs grinded and aqueous extract is taken a glassful .continued for 2 – 3 Months / asper the case.
39.	*Mentha spicata*	Lamiaceae	Pudina	Whole plant is emasculated and extracted in water ,taken in early hours and before sleep .
40.	*Musa paradisiaca* L.	Musaceae	Kela	Stem extract reduces diabetics.
41.	*Murraya koeingii* (L.)	Rutaceae	Curry patta	Leaf juice is taken to treat diabetics.
42.	Morinda citrifolia L.	Rubiaceae.	Noni	Flowers and unripe fruits mixed with 100 ml of cow milk and dose was continued for a month / as per the case.
43.	Murraya koenigii (L.)	Rutaceae	Curry patta	Fresh leaves juice taken directly, used in regular diet
44.	*Ocimum santum* L.	Lamiaceae	Tulsi	Leaf powder with honey is taken orally
45.	*Phyllanthus emblica L.*	Phyllanthaceae	Amla	A paste (100g) made out of the boiled unripe fruit and equal quantity of fine rice, given with normal meal
46.	*Piper nigrum L.*	Piperaceae.	Kaali mirch	The whole plant is made into paste. About 15g of paste given with old jaggery twice a day to check the excessive urination.
47.	*Sida cordata (Burm.f.)*	Malvaceae	Farid buti	Leaf juice (20 mL) mixed with 1-2 spoons honey given once daily
48.	*Syzygium alternifolium*	Myrtaceae	Jamun	Fruits taken directly.However, the seed powder (4-6 gm) given twice daily is more effective than the fruit.
49.	*Trigonella foenum-graecum L*	Leguminosae.	Kasoori methi	. The seed was crushed the juice was taken as tonic with vegetable juice.
50	*Ventilago maderaspatana Gaertn.*	Rhamnaceae	Keonti	Seeds used in curries ,taken directly with milk or water.
51.	Vinca rosea	Apocynaceae	Sadabahar	Leaf juice is taken orally to treat diabetes.
52.	Withinia somnifera	Solanaceae	Ashwagandha	The juice of the leaves used for diabetes
53.	Zingiber officinale Roscoe	Zingiberaceae	Adrak	Rhizomes dried and powdered mixed in curries ,soups in little proportions , taken with milk with empty stomach

ACKNOWLEDGEMENT

The authors are cordially grateful to the people inhabiting in different localities of Bishnugarh block area of Hazaribagh because of their kind support and co-operation during the field surveys.

REFERENCE

1. **Babu, P.A., Suneetha, G., Boddepalli, R., Lakshmi, V.V., Rani, T.S., Rambabu, Y., Srinivas, K. 2006.** A database of 389 medicinal plants for diabetes. *Bioinformation* **1(4):**130-171.
2. **Grover, J.K., Vats, V. 2001.** Shifting Paradigm from conventional to alternate medicine. An introduction on traditional Indian medicine.*Asia Pacific Biotechno News* **5(1)**:28-32.
3. **Chopra, R.N., Nayar, S.L., Chopra, I.C. 1986.** Glossary of Indian Medicinal Plants, Council of Scientific and Industrial Research, ND.India.
4. **Mukherjee, P.K., Maiti, K., Mukherjee, K., Houghton, P.J. 2006.** Leads from Indian medicinal plants with hypoglycemic potentials. *J Ethnopharmacol*. 106(1):1-28.
5. **Farnsworth, N.R. 1998.** Screening plants for new medicines. In Wilson, E.O. (Ed) Biodiversity National Academy Press, , WashingtonDC, pp 83-97.
6. **Ved, D.K., Goraya, G.S. 2008.** *Demand and Supply of Medicinal Plants in India*, Bishen Singh, Mahendra Pal Singh, Dehra Dunand FRLHT, Bangalore, India.
7. **Ragupathy, S., Newmaster, S.G., Maruthakkutti, M., Velusamy, B., Ul-Huda, M.M. 2008.** Consensus of the 'Malasars' traditional aboriginal knowledge of medicinal plants in the Velliangiri holy hills, India. J Ethnobiol Ethnomed. 27(4):8-15.
8. **Pranav Kumar Prabhakar, Mukesh Doble, 2011**. Mechanism of action of natural products used in the treatment of diabetes mellitus In *Chinese Journal of Integrative Medicine* **17(8)** 563-574
9. **Mohan, V. 2004**. Why Are Indians More Prone to Diabetes? *Journal of the Association of Physicians of India* **52,** 468
10. **Grover, J.K., Yadav, S., Vats, V. 2002**. Medicinal plants of India with antidiabetic potential. *Journal of Ethno pharmacology* 81-100.
11. **Shanmugasundaram, E.R., Gopinath, K.L., Shanmugasundaram, K.R. 1990.** Possible regeneration of the islets of Langerhans in streptozotocin-diabetic rats given *Gymnema sylvestre* leaf extracts. *J Ethnopharmacol* **30:**265-279.
12. **K. Kadhirvel** Investigations on Anti-Diabetic Medicinal Plants *Ethnobotanical Leaflets* **14:** 236-47, 2010.
13. **M. Pavani *et al.* 2012**: Ethnobotinical Explorations on Anti Diabetic Plants. *Indian Journal of Fundamental and Applied Life Sciences* Vol. **2 (3)**, pp.100-105/Pavani et al.
14. **Jain, S.K. & Goel, A.K. 1995**. Workshop Exercise-1. Proforma for Field Work, 142-147. *In:* Jain, S.K. (ed.). *A Manual of Ethnobotany* (Scientific Publicatons Jodhpur).

* * *